

Author Response for Submission “PushAround: Collaborative Path Clearing via Physics-Informed Hybrid Search”

Author Names Omitted for Anonymous Review.

To Associate Editor

Q: The reviewers acknowledge the contribution of the paper, however, they have some concerns that need to be addressed.

A: We sincerely thank the Associate Editor for the careful handling of the submission and for summarizing the reviewers’ constructive comments. The revised draft addresses every point raised by Reviewers 3 and 4 through clarification of the WCCG connectivity interpretation, the role and guarantees of the hybrid search objective, the hardware execution interface, state-based online replanning, and the comparison protocol. It also introduces dedicated validation for physics-model mismatch, WCCG connectivity, density statistics, and scaling with the number of robots. Numerical quantities that require new experiments are marked in red with `\todo{}` in this initial response draft.

The principal revisions are summarized below.

- Physics-model robustness. A new planner–execution mismatch study perturbs mass, friction, robot force limits, and contact dynamics while keeping the planning model nominal. The main manuscript and Supplementary Material now contain the protocol and result placeholders. **Insert final perturbation levels, trial count, and numerical conclusions.**
- WCCG connectivity justification. Sec. III-A now interprets WCCG through $W/2$ configuration-space inflation and explicitly discusses non-convex boundary decomposition. A new connectivity-validation experiment compares WCCG with an independent rasterized configuration-space flood-fill reference. **Insert final agreement/false-positive/false-negative statistics.**
- Search objective and guarantees. Eq. (5) is reformulated as a clearance cost used to guide the search. The revised text states explicitly that PIHS uses sampled

actions, bounded lookahead, a heuristic priority, and first-feasible termination; hence no global optimality or completeness claim is made for the underlying continuous hybrid problem.

- Hardware force/control interface and coupled contacts. The revision explains that optimized contact forces are feasibility variables rather than direct hardware setpoints. The Mecanum robots execute planar velocity commands while tracking mode-defined contact poses. Short-horizon simulation propagates all movable bodies, so contact-induced motion of neighboring objects is retained in the successor state.
- Execution discrepancy and replanning. The previous phrase “deviation from simulation” is replaced by explicit state-based monitoring of pre-contact reachability, contact tracking/stall, task predicates, and updated WCCG/schedule validity. Motion capture provides the required robot/object states in the hardware experiments.
- Experimental protocol and statistics. The manuscript now states shared reachability/collision checks and computation budgets. The density suite correctly records 16 workers, an 800 s timeout, and 20,000 control steps. The revised density experiment increases the number of independent seeds and reports dispersion. **Insert final seed count and regenerated statistics/figure.**
- Robot-team scaling. The complexity discussion now explains the N -dependence of contact replacement and the $2N$ -dimensional force LP, and a new simulation table varies $N = 1, 2, 3, 4$. **Insert final robot-scaling data.**

The detailed responses below provide the corresponding manuscript locations and revised text in blue. Added experimental values and conclusions that remain to be collected are shown in red.

To Reviewer 3

Q: Overall, this is a good contribution to the field and a well-written paper. The combination of geometric reasoning, recursive pushing, and physics-informed search is interesting, and the real-world validation strengthens the contribution.

A: We sincerely thank the reviewer for the positive assessment of PushAround and, in particular, for recognizing the end-to-end integration, recursive pushing, dense-clutter scalability, and hardware validation. The revision focuses on the reviewer’s two stated weaknesses and four follow-up questions without changing the central planning framework.

Q: Physics-model robustness is insufficiently characterized: The method relies heavily on simulation-based feasibility validation, but sensitivity to uncertainty in mass, friction, force limits, contact dynamics etc. is not systematically evaluated.

A: Thank you for identifying this important limitation. We agree that the original hardware experiments demonstrate closed-loop recovery from execution error but do not isolate sensitivity to individual physics-model parameters. The revised draft therefore adds a dedicated planner–execution mismatch experiment.

- Mismatch rather than retuning. Planning always uses the nominal mass, friction, force-limit, and contact model. Only the execution simulator is perturbed, so the test measures sensitivity to an incorrect planning model.
- Perturbation families. Obstacle mass, robot–object friction, available pushing force, and contact dynamics are varied separately using the same scene seeds. **Insert final perturbation levels and number of trials.**
- Metrics. The revision reports success rate, execution time, number of replans, and push count. This distinguishes outright failures from cases where online replanning absorbs moderate mismatch.
- Claim scope. The manuscript no longer relies on the hardware runs alone to imply general model robustness; it explicitly states that simulation validation is model dependent and quantifies the sensitivity.

Changes in the revised manuscript:

- The following statement was added in Sec. I-B to make the model-dependence explicit: *Because this physical validation is model dependent, the revised experiments explicitly quantify sensitivity to planner–execution mismatch in mass, friction, available force, and contact dynamics.*
- A new subsection, Sec. IV-A5, was added with the following protocol: *Because PIHS validates candidates with a physics simulator, robustness is evaluated under deliberate planner–execution model mismatch. The planner retains the nominal model used to generate the schedule, while the execution simulator perturbs one parameter family at a time: obstacle mass, robot–object friction, available robot force, and contact dynamics. This protocol distinguishes sensitivity to model error from retuning the planner to the perturbed model.*
- A result figure and the corresponding discussion have been placed in the main paper, and the full table/protocol are included in the Supplementary Material. *Insert final numerical result and robustness conclusion.*

Q: Limited evaluation of more complex scenarios: Although the method is tested with up to 45 movable obstacles, the number of independent trials is limited.

A: We agree. The original density sweep used five independent seed sequences per obstacle density, which is too small for a strong statistical statement at the highest clutter levels. The revised experiment expands this set to *insert final number, target at least 10–20* independent seed sequences. Within one seed, increasing obstacle counts remain nested by adding objects, but different seeds are independently generated. The revised figure reports success and timing dispersion rather than means alone.

Changes in the revised manuscript:

- Sec. IV-A6 now states: *The revised evaluation expands the original five seeds to insert final number, target at least 10–20 independent seed sequences per density. Within each seed sequence, denser instances are generated by incrementally adding objects, while different seeds define independent base layouts.*
- The density figure caption now explicitly requires error bars/dispersion: *The revised plot reports success rate and planning/execution time with dispersion across independent seeds. Replace the current five-seed figure by the expanded-seed mean±standard-deviation/confidence-interval plot.*

Q: The experiments primarily use two robots. What happens when the number of robots increases with the complexity of the task? How does the search and the contact/force formulation scale to larger robot teams?

A: Thank you for raising the team-size question. The revised paper addresses it in both the complexity analysis and a dedicated simulation study.

The key point is that the implementation does not enumerate all joint contact tuples. If each robot has $|\mathcal{A}|$ reachable contact candidates, exhaustive assignment would require $|\mathcal{A}|^N$ tuples. Instead, one greedy replacement sweep tests at most $N|\mathcal{A}|$ single-contact substitutions. Each test solves the force LP whose decision vector has dimension $2N$. Thus larger teams can enlarge the feasible wrench set but also increase contact assignment and LP cost. The number of strategies entering simulation remains bounded by the retained-mode and per-expansion candidate caps.

A new table varies $N \in \{1, 2, 3, 4\}$ and reports success, mode-generation time, planning time, and push count. **Insert final results and interpretation.**

Changes in the revised manuscript:

- Sec. III-E2 now adds: **If $|\mathcal{A}|$ reachable contact candidates are considered per robot, one greedy replacement sweep tests at most $N|\mathcal{A}|$ single-contact substitutions rather than enumerating $|\mathcal{A}|^N$ joint tuples. Each substitution solves the LP in Eq. (10), whose force-vector dimension is $2N$.**
- Sec. IV-A7 and Table **final table number** provide the dedicated robot-team scaling experiment. **Insert final numerical data.**

Q: What guarantees, if any, does the hybrid search provide regarding solution quality?

A: We agree that this should have been stated explicitly. No global optimality or completeness guarantee is intended for the underlying continuous hybrid problem. PIHS samples a finite set of pushing directions/contact modes, uses bounded gap lookahead and a heuristic best-first priority, and terminates at the first physics-validated W -clear solution. Moreover, the remaining-cost term is not required to be admissible. Equation (5) therefore defines the clearance cost used to rank schedules rather than an optimization problem that PIHS claims to solve globally.

Changes in the revised manuscript:

- Eq. (5) is now introduced as the clearance objective $\mathcal{J}_{\text{clr}}(\pi)$ rather than as an implicit global-minimization claim.
- Sec. III-D now states: **PIHS is a feasibility-oriented heuristic search. Equation (5) defines the schedule cost used to rank candidates, while the implemented search samples a finite set of pushing directions and modes, uses bounded gap lookahead, and terminates when a feasible W -clear schedule is found. Therefore, no**

global optimality or completeness claim is made with respect to the underlying continuous hybrid planning problem.

- The termination statement was revised to say that the returned schedule is a feasible solution without implying global minimization of $\mathcal{J}_{\text{clr}}$.

Q: The objective includes the total duration T and the control-effort cost J . Is the time needed for the final vehicle to actually traverse the cleared corridor included in this objective, or is the objective only concerned with clearing the obstacles? Similarly, does J only capture the robots' effort in moving the obstacles?

A: The objective concerns the robotic clearance operation only. The external vehicle traversal time is not included in T ; vehicle passage is a terminal feasibility condition. Similarly, J represents the robots' manipulation/control effort during obstacle pushing. We have made both points explicit to remove the ambiguity.

Changes in the revised manuscript:

- Sec. II-D now states: $T \triangleq \sum_{k=1}^K \Delta t_k$ is the duration of the robotic clearance operation only; the subsequent traversal time of the external vehicle is not included. The weight $\alpha > 0$ trades time against $J(\cdot)$, which measures the robots' simulation-estimated control/manipulation effort for moving obstacles. Vehicle traversal enters the formulation only through the terminal W -clearance condition.

Q: The paper demonstrates online replanning when the real execution differs from the simulation, How can simple pushing robots evaluate this discrepancy?

A: Thank you for pointing out that the original wording could be interpreted as requiring online parameter identification or force sensing. That is not the implementation. The phrase "deviates from simulation" has been replaced by an explicit state-based execution monitor.

In the hardware experiments, motion capture supplies robot and obstacle poses. The controller checks whether the pre-contact transition is feasible, whether robots maintain/track their intended contact poses, whether motion stalls, whether the task-specific geometric push predicate is achieved, and whether the updated WCCG still supports the remaining schedule. A failure of any of these checks triggers replanning from the observed state. Thus simple velocity-controlled pushers do not need to estimate the underlying mass/friction error itself.

Changes in the revised manuscript:

- Sec. III-E1 now states: Replanning does not require the robots to estimate an unknown force-model error directly. The execution monitor uses observed robot/object poses and task predicates: replanning is triggered when a pre-contact transition is blocked, contact tracking stalls or loses the planned contact, the pushed object fails to satisfy the corresponding geometric task predicate within its rollout budget, or the observed post-push geometry invalidates the remaining schedule/WCCG connectivity.
- The Supplementary Material reports the current contact-position threshold (0.10 m) and stall check (less than 0.05 m motion over 0.25 s), together with the task-predicate logic.

To Reviewer 4

Q: Overall, I think the approach is technically reasonable and the experiments support the proposed method. My comments are mainly about clarification rather than major technical concerns, and I would support acceptance after these revisions.

A: We sincerely thank the reviewer for the positive assessment and for identifying four places where the original presentation was insufficiently explicit. The revision addresses each point directly, as detailed below.

Q: First, the WCCG connectivity condition is important since it is used throughout the search and also determines termination. The intuition is reasonably clear, but I think the paper would benefit from a little more justification here, or perhaps a comparison with a standard configuration-space connectivity check. This seems especially relevant when the obstacles are non-convex.

A: We agree. The revised Sec. III-A now makes the configuration-space interpretation explicit and adds the suggested independent comparison.

For a vehicle represented by a radius- $W/2$ enclosing disk, inflating obstacle geometry by $W/2$ converts every pair of boundary components separated by less than W into overlapping clearance barriers. WCCG bridge-bridge edges encode these narrow barriers. Non-convex obstacles are first decomposed into clearance-relevant convex components/boundary segments before the closest-point bridges are inserted. The same-face condition is therefore interpreted as a topological query on the planar regions not separated by these inflated barriers.

Because implementation details and degenerate non-convex cases still matter, we also add an empirical check against a conventional rasterized configuration-space flood-fill reference, including dedicated non-convex scenes. **Insert final trial count, resolution, agreement, and false-positive/false-negative results.**

Changes in the revised manuscript:

- Sec. III-A2 now adds: **The geometric interpretation follows the standard configuration-space inflation for the vehicle-center point. Expanding every obstacle boundary by a disk of radius $W/2$ transforms a pair of boundary components whose closest distance is below W into overlapping clearance barriers. The corresponding WCCG bridge-bridge edge records this non-traversable bottleneck.**
- The previous statement that WCCG “eliminates resolution-induced errors” was softened. The revised remark states that WCCG avoids choosing a workspace grid resolution but still depends on correct decomposition/bridge construction.

- Sec. IV-A3 and Supplementary Sec. S2 add the requested C-space comparison.
Insert final result.

Q: Second, I was a little confused about the relationship between Eq. (5) and the actual search procedure. Eq. (5) is written as an optimization problem, while the implementation uses sampled pushing directions, limited lookahead, and a heuristic best-first search that terminates once a feasible solution is found. The authors should clarify whether any optimality or completeness claim is intended. If not, it may be better to describe Eq. (5) as the objective used to guide the search.

A: We agree completely with this interpretation and have revised the formulation accordingly. Eq. (5) now defines the clearance cost $\mathcal{J}_{\text{clr}}(\pi)$ used to guide candidate ranking. PIHS is explicitly described as a feasibility-oriented heuristic search with finite sampling, bounded lookahead, non-admissible heuristic ranking, and first-feasible termination. No global optimality or completeness claim is intended.

Changes in the revised manuscript:

- Sec. II-D replaces the minimization display by the objective definition: $\mathcal{J}_{\text{clr}}(\pi) \triangleq T + \alpha \sum_{k=1}^K J(m_k, \xi_k, \mathbf{s}(t_k))$.
- Sec. III-D explicitly states the absence of global optimality/completeness guarantees and notes that the remaining-cost term is not required to be admissible.

Q: Third, it is not completely clear how the optimized contact forces are used on the hardware. The pushing mode includes both contact points and forces, but the execution section mainly describes moving the robots to the contact configuration and then following the desired pushing direction. It would be useful to briefly explain the low-level controller and whether these forces are actually commanded or are mainly used to check feasibility. Related to this, I am also not sure whether coupled motion between multiple objects, such as in the second hardware experiment, is explicitly considered during planning or is mostly handled by the physics simulation.

A: Thank you for identifying both ambiguities. The optimized forces are primarily planning-level feasibility variables; they are not sent to the hardware as force commands. For a candidate contact tuple, the LP determines whether the robots can generate a wrench compatible with the desired object motion within the assumed force/friction constraints. The selected hardware mode is then executed by closed-loop planar velocity control. Specifically, the Mecanum UGVs receive ROS geometry_msgs/Twist

commands while tracking the assigned contact poses and a short forward object-motion reference using motion-capture feedback.

Coupled object motion is represented through simulation. Although a high-level strategy names one directly targeted obstacle, *EvalSim* advances all robots and all movable bodies. Object-object contacts are therefore resolved by the physics engine, and the complete resulting scene state is returned to the search. The adjacent T-shaped object moved in hardware Scenario 2 is exactly this type of contact-induced motion rather than a separately scripted task.

Changes in the revised manuscript:

- Sec. III-C2 now states: The optimized force vector is used here to certify and rank wrench feasibility under the modeled force limits. On the hardware, the robots execute pose/velocity commands that maintain these contact locations rather than directly commanding the optimized forces.
- Sec. III-E1 now specifies the hardware interface: The controller converts the planned contact geometry and desired object motion into planar robot velocity commands (ROS Twist); motion-capture poses provide the feedback used to track the moving contact poses.
- Sec. III-C3 now clarifies coupled objects: The rollout advances all robots and all movable bodies jointly in the physics engine. Consequently, object-object contacts induced by pushing the target obstacle are resolved during *EvalSim* and their coupled motion is retained in the full successor state.

Q: The experiments are generally convincing. The paper already mentions that the baselines use the same W-clearance criterion, object distribution, and execution simulator. However, a few more implementation details, such as robot reachability checks, timeout limits, and computation budget, would make the comparison easier to interpret. I would also suggest adding error bars or standard deviations, particularly for the scalability experiment where only five seeds are used for each obstacle density.

A: We agree and have expanded both the main setup and the Supplementary protocol.

- Reachability. A contact candidate is retained only when the associated pre-contact pose is collision free and the transition planner can reach it from the current robot configuration. A multi-robot tuple is rejected if any assigned robot fails this check.
- Common budgets. The main 40-trial comparison now includes the shared timeout, search/control cap, and parallel-worker count. Verify and insert the exact Scenario-1 values.

- Density-suite budgets. The stored evaluation data specify 16 parallel workers, an 800 s per-trial timeout, and a 20,000-control-step cap. The original manuscript’s 400 s statement is corrected to 800 s. Failed and pruned trials are assigned this same cap in the planning-time statistic.
- Statistics. The density sweep is expanded from five seeds to **insert final count**, and the regenerated figure reports variability (mean $\pm$ standard deviation or confidence interval) rather than means alone.

Changes in the revised manuscript:

- Sec. III-C2 adds the explicit pre-contact reachability criterion.
- Sec. IV-A1 now states: **All compared methods use the same robot footprints, obstacle geometry, collision/reachability checks, success criterion, and execution simulator. A candidate that requires an unreachable or collision-infeasible pre-contact robot configuration is rejected before pushing.**
- Sec. IV-A6 now states the density-suite computation budget and replaces the original 400 s cap by the correct 800 s timeout. The figure caption also requires the expanded-seed error-bar plot.